Science Practice Test — Forces and Magnets

Year 3 · AV3 · 3rd term (practice)

Name:	Date:	Class:	Score:

Practice test made from the school's AV3 study guide. It is not an official school test.

Read each question to the end. Use a pencil. In the writing questions, answer with complete sentences.

Block 1 of 3 · about 10 minutes · then a break

Part A — Forces

2 points

Circle the correct answer.

1. What is a **force**?
   a. a kind of light
   b. a push or a pull
   c. a kind of sound

2. Which force makes an apple **fall down** from a tree?
   a. gravity
   b. friction
   c. magnetic force

3. Which force makes a rolling ball **slow down** on the grass?
   a. gravity
   b. electric force
   c. friction

4. Which force works **without touching**?
   a. a push with your hand
   b. friction
   c. gravity

5. Why is magnetic force a **non-contact** force?
   a. It needs to touch first.
   b. It pulls without touching.
   c. It only works on wood.

6. Which force pulls bits of paper to a rubbed balloon?
    a. electric force
    b. friction on paper
    c. gravity from below

7. You jump up. Why do you **come back down**?
    a. A magnet pulls me.
    b. Friction pushes me.
    c. Gravity pulls me down.

8. What does friction do to a moving thing?
    a. It makes it go faster.
    b. It slows it down.
    c. It makes it glow.

Part B — Contact or non-contact?

0.8 points

Put a tick in the correct box for each force.

	contact	non-contact
friction		
gravity		
a kick		
magnetic force		
a push with your hand		
static electricity		

Part C — Write

0.9 points

Answer with complete sentences.

9. Explain how a **compass** works. Why does the needle point north?

10. What is **friction**? Give one example of how it makes movement slower.

Break: stand up, drink some water and come back.

Block 2 of 3 · about 10 minutes · then a break

Part D — True or false

1 point

Write T (true) or F (false). If it is false, write the correct sentence.

11. ( ) A magnet can pull a wooden spoon.
    Correct it: _____

12. ( ) Different poles of two magnets attract each other.
    Correct it: _____

13. ( ) Two magnets with the same poles facing each other attract.
    Correct it: _____

14. ( ) The magnetic field is the area around a magnet where its force works.
    Correct it: _____

15. ( ) A compass needle is a tiny magnet.
    Correct it: _____

16. ( ) A temporary magnet works forever.
    Correct it: _____

Part E — Think about it

1.2 points

Circle the correct answer.

17. Two magnets push apart. Which explanation is right?
    a. The same poles are facing.
    b. Different poles are facing.
    c. The magnets are too old.

18. A nail touches a magnet and picks up a clip. What kind of magnet is the nail?
    a. a permanent magnet
    b. a temporary magnet
    c. an electromagnet

19. You do **not** rub a balloon. What happens when you hold it over paper bits?
    a. It pulls even more paper.
    b. It turns into a magnet.
    c. It does not pull the paper.

20. Which of these has a magnet inside and shows directions?

    a. a ruler
    b. a compass
    c. a calculator

21. You take the magnet away from the nail that was holding a clip. What happens?

    a. The nail stops holding the clip.
    b. The nail holds the clip harder.
    c. The clip turns into a magnet.

22. Which object at home uses a magnet?

    a. a soft pillow
    b. the fridge door
    c. a paper bag

Part F — Short answers

0.8 points

Write one word or a short phrase.

23. What are the two poles of a magnet called?

_____

24. Name one thing that static electricity can do (use a balloon in your example).

_____

25. Name one everyday object that has a magnet or is pulled by one.

_____

26. Static electricity can be dangerous. What is a giant static spark in the sky?

_____

Break: stand up, drink some water and come back.

Block 3 of 3 · about 10 minutes · last block

Part G — Words and examples

1 point

Write the number of the word next to its example.

1. gravity
2. friction
3. electric force
4. magnetic force
5. electromagnet

( ) a. a magnet pulls a nail

( ) b. a box slows down when you push it on the floor

( ) c. a magnet that works when electricity is on

( ) d. a ball falls back to the ground

( ) e. a rubbed balloon pulls little bits of paper

Part H — Does the magnet pull it?

0.8 points

Put a tick in the correct box for each thing.

	pulled by a magnet	**not pulled**
paper clip		
eraser		
steel nail		
glass cup		
steel key		
wooden pencil		

Part I — Complete the sentences

1 point

Use the words in the word bank. Two words are extra.

Word bank: attract · repel · north · field · rub · melt

27. Different poles _____ each other.

28. Same poles _____ each other.

29. A compass needle points _____ because the Earth acts like a giant magnet.

30. A clip that is inside the magnetic _____ is pulled by the magnet.

31. Friction happens when two surfaces _____ together.

Part J — Draw

0.5 points

Draw and label with pencil.

Draw two bar magnets that **attract**. Write N and S on each one.

Draw two bar magnets that **repel**. Write N and S on each one.

Good luck! · Total: 10

Answer key (Gabarito) — Science Practice Test — Forces and Magnets

> Para quem aplica: a prova vem em 3 blocos curtos, com pausa. Leia a instrução com ele antes de começar cada bloco; deixe-o se levantar e se mexer entre os blocos.

Block 1

Part A — Forces

1. b) a push or a pull
2. a) gravity
3. c) friction
4. c) gravity
5. b) It pulls without touching.
6. a) electric force
7. c) Gravity pulls me down.
8. b) It slows it down.

Part B — Contact or non-contact?

friction: contact
gravity: non-contact
a kick: contact
magnetic force: non-contact
a push with your hand: contact
static electricity: non-contact

Part C — Write

9. **Expected answer (example):** A compass needle is a tiny magnet. The Earth is like a giant magnet, and its magnetic field turns the needle to point north.
   Para o responsável, confira: a agulha é um ímã; a Terra age como um ímã gigante / campo magnético da Terra (0,5 no total)
10. **Expected answer (example):** Friction is a force between two surfaces that rub together. A ball stops sooner on grass than on a smooth floor.
    Para o responsável, confira: superfícies que se esfregam; exemplo de algo que fica mais lento (0,4 no total)

Block 2

Part D — True or false

11. F (false) — A magnet pulls iron and steel things, not wood.
12. T (true)
13. F (false) — Same poles repel (push apart).
14. T (true)
15. T (true)
16. F (false) — A temporary magnet works only for a while.

Part E — Think about it

17. a) The same poles are facing.
18. b) a temporary magnet
19. c) It does not pull the paper.
20. b) a compass
21. a) The nail stops holding the clip.
22. b) the fridge door

Part F — Short answers

23. north (N) and south (S)
24. pull little bits of paper (or: a rubbed balloon sticks to the wall)
25. fridge door, cupboard catch, compass, pencil case, paper clip, steel key (one is enough)
26. lightning

Block 3

Part G — Words and examples

a → 4 · b → 2 · c → 5 · d → 1 · e → 3

Part H — Does the magnet pull it?

paper clip: pulled by a magnet
eraser: not pulled
steel nail: pulled by a magnet
glass cup: not pulled
steel key: pulled by a magnet
wooden pencil: not pulled

Part I — Complete the sentences

27. attract
28. repel
29. north
30. field
31. rub

Part J — Draw

Expected answer (example): N e S de um frente a S e N do outro (polos diferentes se encaram).
Para o responsável, confira: dois ímãs; N e S escritos em cada um; polos diferentes de frente um para o outro (0,25)

Expected answer (example): N frente a N, ou S frente a S.
Para o responsável, confira: dois ímãs; N e S escritos em cada um; polos iguais de frente um para o outro (0,25)